

Fish Classification Using Frequency Response Data Final Report

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1. Introduction

This project aims to classify fish species using sonar-based frequency response data. The dataset consists of measurements taken from two fish species: Lake Trout (LT) and Smallmouth Bass (SMB). Classifying these two species is important for fisheries management, conservation efforts, and ecological monitoring in Ontario's freshwater ecosystems, including the Great Lakes and inland lakes near Toronto.

Lake Trout populations have historically declined due to overfishing, invasive species, and climate change, while Smallmouth Bass are expanding their range, potentially disrupting native Lake Trout habitats. Therefore, understanding their distributions is critical for monitoring ecosystem shifts and guiding conservation strategies.

2. Data Section

The dataset contains 10,107 observations and 276 variables after cleaning, where repeated frequency tests were conducted per fish. By placing a fish underwater, the frequency data was collected using a sonar system, where sound waves were transmitted from a boat, bounced off a fish, and received back.

The key features include:

- **Fish ID:** Unique identifier for each fish. (fish_Num)
- **Species:** The categorical variable representing Lake Trout or Smallmouth Bass (species)
- **Frequency Variables (F45-F260):** Numeric values representing the frequency response measurements. (F45, F45.5,...,F260)
- **Biological Features:** Total length, fork length, weight, girth, dorso-lateral height, sex, air bladder total length, air bladder width, and air bladder weight.

Data Preprocessing

The original dataset has 14,575 observations and 484 variables. It includes three fish species (Lake Trout, Lake Whitefish, and Smallmouth Bass), time-related data (e.g., dateTimeSample, dateProcessed, Ping_time), and spatial data (e.g., Target_depth_mean, Target_range). However, since I am only using Lake Trout and Smallmouth Bass for classification, I excluded Lake Whitefish. Additionally, because the fish were manually placed underwater, the spatial data appears artificial. Therefore, after feature selection, I created the following two datasets:

- **fish_data:** Includes biological features alongside frequency response data to analyze their relationship.
- **freq_only:** Contains only fish labels and frequency responses to focus on species-specific patterns.

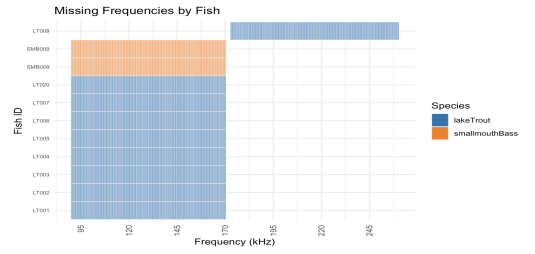


Figure 1. Missing frequency per fish Heatmap

After feature selection, I counted the missing values for each fish and identified which frequency ranges were missing using a heatmap (Fig. 1). The heatmap clearly shows that most fish are missing data in the 90–170 kHz frequency range, while only the fish with ID LT008 has data missing in the 173–260 kHz range, so I did the following action for handling the missing values:

- The 90–170 kHz frequency range was removed due to excessive noise and inconsistencies.
- The fish with ID LT008 was removed since it had over 50% missing values.

I also checked all variable types and found that the Frequency Response and Sex variables were stored as characters. I converted them to numeric for easier use in the analysis.

In my Exploratory Data Analysis (EDA) step:

1. I first examined the Mean Frequency Response Strength Patterns per Fish group by species and observed that Lake Trout exhibited stronger responses than Smallmouth Bass across most frequency ranges, with clearer separation in the 45–90 kHz range.
2. For the T-Test Analysis, after reviewing my teacher’s comment on my EDA report regarding Bonferroni correction and confirming that the frequencies are correlated, I decided not to use the T-test analysis.
3. I also checked for outliers in the Frequency Response variable but found that the difference in mean strength per frequency per fish did not vary significantly. Figure 2 shows that most differences are close to 0, with only a few ranging between 2 and 3. Therefore, I decided not to remove these outliers

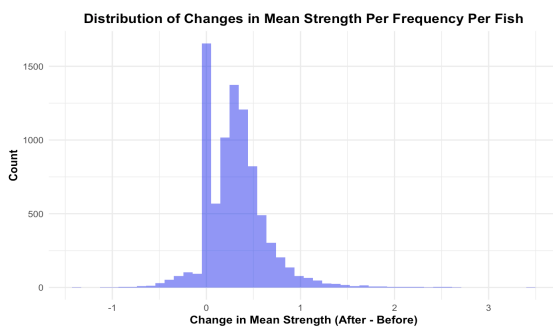


Figure 2. Histogram for distribution of differences in mean strength

- Then, I used Principal Component Analysis (PCA). From Figures 3 and 4, we observe that the most important frequencies fall within the 63–77 kHz range, contributing significantly to PC1, which explains 48.6% of the total variance. These frequencies likely capture broad variations in fish frequency responses, but their role in distinguishing species remains uncertain. However, I will keep the PCA frequency range in the modeling step for exploratory purposes.

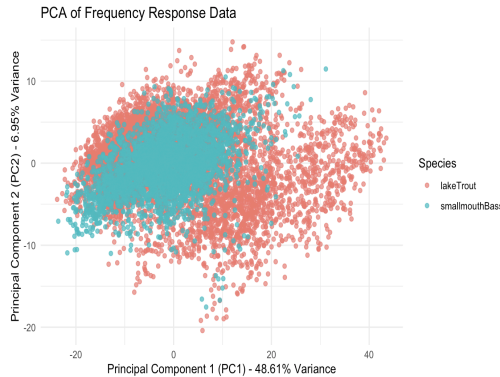


Figure 3. PCA scatter plot

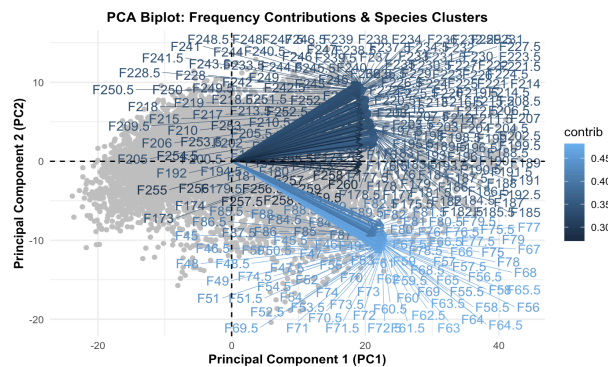


Figure 4. PCA Biplot

- I also conducted a Correlation Analysis. Figure 5 shows that adjacent frequencies exhibit strong correlations, highlighting redundancy in some signals. Figure 6 illustrates that correlations between biological features and frequency responses vary by species, with Lake Trout responses being more influenced by air bladder width and Smallmouth Bass by body size. Therefore, I can use key biological features that vary by species in the modeling process.

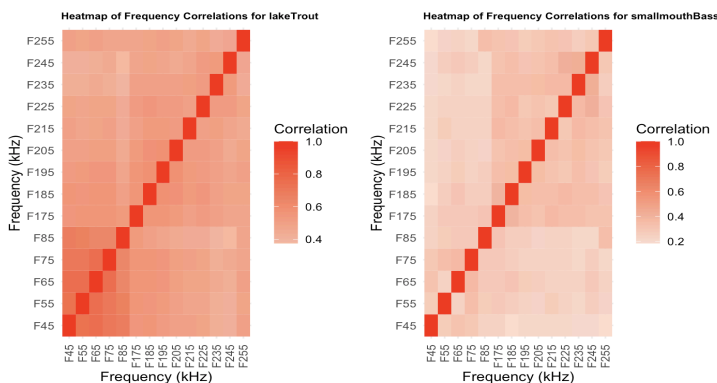


Figure 5. Correlation between frequency heatmap

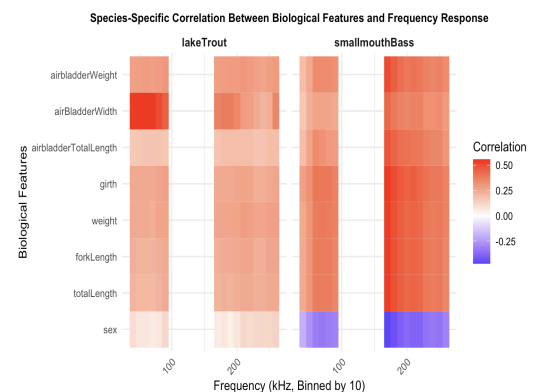


Figure 6. Biological Feature VS. Frequency Heatmap

3. Method Section

Data Preparation:

To evaluate the impact of different features on classification performance, I constructed six variations of the dataset. Each dataset includes different combinations of biological features and frequency response data:

- All Biological Features + All Frequency Response Data**

- Includes all available biological measurements (e.g., length, weight, girth, air bladder characteristics) along with frequency response features.

2. Only Frequency Response Data

- Excludes biological features and relies solely on frequency response data for classification.

3. Only PCA-Selected Frequency Range

- Excludes biological features and uses only the PCA-selected subset of frequency response data.

4. PCA-Selected Frequency Range + Species-Specific Biological Features

- Uses PCA-determined frequency features while selecting the top 3 biological features most strongly correlated with frequency response for each species.

5. Species-Specific Biological Features + All Frequency Response Data

- Uses the top 3 biological features most strongly correlated with frequency response for each species while keeping all frequency response data. For example, some species may have additional measurements like girth and sex, while others may not.

6. PCA-Selected Frequency Range + All Biological Features

- Applies Principal Component Analysis (PCA) to identify key frequency components, combined with all biological features.

Next, to address the research question of classifying fish species based on sonar frequency response data, I first implemented a custom data splitting strategy to avoid data leakage and preserve species balance. Specifically, I split the dataset into 80% training and 20% test sets based on unique fish IDs, ensuring that all observations from the same fish appear in only one of the two sets. This helps maintain the independence of training and testing data while controlling for within-fish correlation.

For modeling, I employ the SuperLearner ensemble method as the primary approach. Given the relatively small sample size ($\leq 10,000$ observations), SuperLearner is well-suited as it allows multiple base models to be run simultaneously, leveraging their combined strengths to improve accuracy while mitigating overfitting. Instead of selecting a single model a priori, SuperLearner uses 5-fold cross-validation to determine the optimal weighted combination of base models, ensuring better generalization to unseen data.

The characteristics of the dataset drive the choice of base learners. I select the following models for inclusion in the ensemble:

- SL.glmnet (Elastic Net Regression): Suitable for high-dimensional data with correlated features, such as frequency response values at different kHz levels.
- SL.randomForest (Random Forest): Captures non-linear interactions between biological and

frequency-based features while handling missing data effectively.

- SL.xgboost (Extreme Gradient Boosting): A powerful tree-based model that optimizes feature selection and provides robustness against noise.
- SL.nnet (Neural Networks): Learns complex hierarchical patterns from the data, useful when subtle variations in sonar responses differentiate species.
- SL.ranger (Fast Random Forest Implementation): Optimized for efficient handling of large feature spaces, which is crucial given the number of frequency response variables.

I use a binomial family model (`family = binomial()`) because the classification task involves distinguishing between two species: Lake Trout and Smallmouth Bass. To train the SuperLearner ensemble, I combine five base learners: GLMnet, Random Forest, XGBoost, Neural Network, and Ranger. The ensemble weights are optimized using a meta-learning method, specified through the `meta_method` argument. In this setup, I use `"method.AUC"` to directly optimize the ensemble based on AUC performance, which is particularly helpful when dealing with class imbalance or overlapping distributions. In general, I select the base learner with the highest ensemble weight and evaluate its performance separately. In cases where SuperLearner assigns roughly equal weights across learners, I default to XGBoost due to its strong past performance and robustness to correlated, non-linear features without strict distributional assumptions.

To evaluate the trained model's performance, I used multiple classification metrics, including accuracy, balanced accuracy, AUC-ROC score, confusion matrix, and F1 score. The model predictions were compared against the true labels from the test dataset, and a confusion matrix was generated to analyze misclassifications and identify whether one class was consistently harder to predict. Since the dataset may have class imbalances, balanced accuracy was calculated as the average of sensitivity (true positive rate) and specificity (true negative rate), ensuring a fair evaluation across both classes. The AUC-ROC score was also computed to assess the model's ability to distinguish between fish species. A higher AUC value indicates stronger classification performance. Additionally, the F1 score, which combines precision and recall, was used to provide a more informative metric when the costs of false positives and false negatives may differ.

4. Results

Table 1 presents the SuperLearner ensemble weights and corresponding cross-validation (CV) risks for each base learner across the six dataset variants. In dataset variants 1, 4, 5, and 6, the ensemble assigned equal weights (0.20) to all five models, indicating that no single learner strongly outperformed the others during cross-validation. This typically reflects either uniform performance across learners or the ensemble's inability to confidently prioritize one model. In contrast, dataset variants 2 (Frequency Only) and 3 (PCA Frequency Only) showed different weighting, with Random Forest and Ranger receiving the highest weights, while GLMNet and Neural Network were excluded (0 weight) due to higher CV risk. For example, in variant 2, SL.ranger and SL.randomForest achieved the lowest CV risks (0.091), while SL.nnet had a notably higher risk (0.496). This suggests that tree-based methods were more robust on frequency-only datasets, especially in the absence of dimensionality reduction or biological features.

Dataset Variant	glmnet W	rf W	xgb W	nnet W	ranger W	glmnet CV	rf CV	xgb CV	nnet CV	ranger CV
1. All Bio + All Freq	0.20	0.20	0.20	0.20	0.20	0.000	0.000	0.000	0.393	0.000
2. Freq Only	0.00	0.377	0.187	0.000	0.436	0.262	0.091	0.101	0.496	0.091
3. PCA Freq Only	0.00	0.488	0.029	0.000	0.483	0.401	0.160	0.197	0.520	0.160
4. Species-Specific Bio + PCA Freq	0.20	0.20	0.20	0.20	0.20	0.000	0.000	0.000	0.329	0.000
5. Species-Specific Bio + All Freq	0.20	0.20	0.20	0.20	0.20	0.000	0.000	0.000	0.380	0.000
6. All Bio + PCA Freq	0.20	0.20	0.20	0.20	0.20	0.000	0.000	0.000	0.494	0.000

Table 1. SuperLearner Weights and CV Risk Summary (W = Wrights, CV = CV Risk, Weight determined by AUC)

Table 2 summarizes the test-set classification performance of the best-performing base model for each of the six dataset variants, selected using both the highest SuperLearner weight and empirical evaluation metrics. The key metrics presented include Balanced Accuracy, AUC-ROC Score, and F1 Score, all of which reflect different aspects of classification effectiveness. The highest-performing models were observed in Dataset 4 (*Species-Specific Bio + PCA Freq*) and Dataset 5 (*Species-Specific Bio + All Freq*), where XGBoost achieved perfect classification results: Balanced Accuracy = 1.00, AUC = 1.00, and F1 Score = 1.00. These results suggest that combining targeted biological features with either the full or PCA-reduced frequency data enabled the model to perfectly distinguish between Lake Trout and Smallmouth Bass. The synergy between tailored biological traits and frequency responses appears to provide highly separable class information.

By contrast, frequency-only datasets, Variant 2 and Variant 3, performed poorly. The best models in these cases (Ranger and Random Forest, respectively) yielded balanced accuracy scores of just 0.41 and 0.47, with relatively low F1 Scores and near-random AUC performance (e.g., AUC = 0.499 in Variant 2). This suggests that frequency response data alone may be insufficient to distinguish species

when not complemented by biological context. Variants 1 and 6, which used all biological features combined with either the full frequency set or PCA-reduced frequencies, resulted in moderate performance. In both cases, XGBoost achieved balanced accuracy scores around 0.63 and F1 Scores near 0.42. While these results outperform frequency-only models, they fall short of the targeted bio-frequency combinations in Variants 4 and 5. This shows the importance of selective feature engineering rather than including all available data arbitrarily. .

Overall, Table 2 illustrates that thoughtful selection and pairing of biological and frequency features substantially improves classification performance, and that even high-capacity models like XGBoost are sensitive to data quality and feature design.

Dataset Variant	Best Model	Accuracy	Balanced Accuracy	AUC-ROC	F1 Score
1. All Bio + All Freq	XGBoost	0.6461	0.6334	1.0000	0.4213
2. Freq Only	Ranger	0.4002	0.4118	0.4990	0.5462
3. PCA Freq Only	Random Forest	0.4606	0.4745	0.5919	0.6114
4. Species-Specific Bio + PCA Freq	XGBoost	1.0000	1.0000	1.0000	1.0000
5. Species-Specific Bio + All Freq	XGBoost	1.0000	1.0000	1.0000	1.0000
6. All Bio + PCA Freq	XGBoost	0.6461	0.6334	1.0000	0.4213

Table 2. Best Model Evaluation per Dataset Variant

Overall, these results highlight the importance of combining dimensionality-reduced acoustic signals with targeted biological features to enhance classification performance. The success of Dataset Variants 4 and 5 demonstrates that careful feature selection, rather than simply using all available data, plays a critical role in building effective predictive models. Moreover, the results confirm the robustness of XGBoost, particularly in handling high-dimensional, non-linear, and potentially noisy data. While the SuperLearner ensemble framework offered a systematic way to evaluate and combine models, in practice, individual base learners, especially XGBoost, delivered excellent and interpretable performance in well-prepared dataset variants. These findings suggest that SuperLearner can serve as

an effective model-selection tool, even when a single learner ultimately dominates the ensemble.

5. Discussion

Result

While the final models demonstrated strong classification performance, it is important to acknowledge a potential limitation related to data leakage. Specifically, species-specific feature selection was conducted using the entire dataset before the train-test split. This means that the top correlated features for each species were selected with access to all observations, including those later used for testing. As a result, the model may have indirectly “seen” patterns from the test set during training, leading to optimistically biased performance estimates. A more robust approach would involve performing feature selection solely within the training data and then evaluating the model on unseen test samples. Incorporating this correction in future work would improve model generalizability and produce a more accurate reflection of real-world performance.

Limitations

This project has several data-related limitations that may have influenced model performance. First, I treated sonar pings as independent observations, without accounting for their temporal order. Since the pings were collected sequentially, modeling the data as a time series could better capture species-specific acoustic patterns. Second, I retained outliers based on visual inspection, as they were not extreme and could carry biologically meaningful signals. However, formal outlier diagnostics may help assess whether they introduce noise or improve model learning. Third, the dataset exhibits class imbalance, particularly at the observation level, where Lake Trout often have more sonar pings than Smallmouth Bass. Although I used balanced accuracy to mitigate bias in evaluation, the imbalance likely affected how the model learned during training. This could lead to overfitting to the majority class. Future improvements may involve resampling techniques or limiting observations per fish to improve class balance.

While the exploratory data analysis provided useful initial insights, some limitations remain. The analysis primarily relied on mean frequency responses, which may have masked variability within individual fish. Incorporating distributional statistics, such as standard deviations or per-fish variance, could offer a more nuanced understanding of within-species variation. Although PCA was used to reduce dimensionality and improve computational efficiency, its effectiveness varied across datasets. Notably, when paired with species-specific biological features, PCA contributed to excellent model performance. However, PCA alone did not yield strong results in frequency-only variants, suggesting that contextual pairing with biological features is key. Additionally, the use of principal components may still obscure some biologically meaningful or frequency-localized signals. The EDA also focused on aggregate patterns and did not explore temporal trends in the sonar responses, which could contain informative sequential structure. Moreover, no formal statistical feature selection techniques (e.g., ANOVA, mutual information) were applied to identify the most discriminative frequencies. Incorporating these techniques could enhance interpretability and guide future feature engineering strategies.

The SuperLearner ensemble, while robust and flexible, is computationally intensive, particularly when applied to high-dimensional frequency data across multiple dataset variations. The time required for training increases significantly as model complexity and dataset size grow, making it less practical for rapid experimentation or large-scale deployment. Although I used 5-fold cross-validation, other

strategies, such as 10-fold or repeated CV which might offer a better balance between computational cost and generalization accuracy. Exploring alternative validation schemes could enhance the ensemble's reliability without significantly increasing training time. A key limitation is that base models were not hyperparameter-tuned. Each model was used with default settings, which may not reflect optimal configurations. Tuning the best-performing models, such as XGBoost or Random Forest, could substantially improve classification metrics like AUC and F1 score. Additionally, some base learners (e.g., Neural Networks) may require scaling or transformation of input features, which wasn't systematically applied in this phase. Inconsistent preprocessing across learners could affect their comparative performance in the ensemble. Lastly, the SuperLearner ensemble is not dynamic: once trained, it does not adapt easily to new data without full retraining. This limits its usefulness in real-time or streaming scenarios unless retraining infrastructure is established.

Future Improvement

Several directions could be explored to improve model performance. First, applying feature selection techniques, either statistically or guided by domain knowledge, could help isolate the most informative frequencies and reduce redundancy. Second, incorporating the time-series structure of sonar pings using sequential models like LSTM or 1D CNNs may reveal temporal patterns that static models miss. Third, hyperparameter tuning of top-performing base learners could lead to significant performance gains, especially in models like XGBoost or Random Forest. Fourth, addressing class imbalance through resampling techniques, such as downsampling, upsampling, or SMOTE, at both the fish and observation level could improve generalization. Fifth, experimenting with alternative ensemble strategies or model stacking approaches might yield further gains over the current SuperLearner setup. Finally, interdisciplinary methods from fields such as psychology, biostatistics, or neuroscience, some of which were introduced in other classmates' presentations, could be adapted. For example, mixed-effects models may help account for repeated measures, and time-frequency analysis techniques could better capture the structure of sonar signals.